

H+c flavour-scheme uncertainty

From a flat 30% to a derived 3FS/4FS shape systematic

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Leading GEN c-hadron p_T · validated on official Run 2 and Run 3 samples

Scope — what this uncertainty covers

This systematic covers ONLY the difference between the 3FS and 4FS flavour schemes.

It does **not** cover the difference between the NLO normalisation of the nominal 4FS FxFx sample and higher-order predictions. LHCHWG-2026-007 (Table 3) finds the NNLO (MiNNLO_{PS}) inclusive H+c cross section **~25% below NLO** at 13.6 TeV.

- Nominal signal is **unchanged**: 4FS FxFx, its own event weights, its NLO cross section
- No stitching, no NNLO normalisation: the sample is NLO, and an NNLO total would not carry the right selection efficiency
- The role of the uncertainty is **identical to Run 2** (AN-23-102 §7.1): *"Obtain the flavor scheme uncertainties (4FS vs. 3FS) ... The differences between the two predictions are used as uncertainty"*
- What changes is **how** that difference is evaluated

Today: a flat 30% inherited from Run 2

item	Run 2 (HIG-24-009, AN-23-102)	this analysis today
nominal	4FS FxFx, $\sigma = 90 \text{ fb}$ (NLO)	4FS FxFx, NLO
FS uncertainty	3FS vs 4FS non-FxFx yield difference	<code>xsec_hplusc_4FS_5FS</code> lnN 1.30
size	"order 30%, 3FS undershooting 4FS"	30%, flat
impact	dominant theory uncertainty; kink in the likelihood	expected limit 921 $\rightarrow$ 1034 (+11%)

Limitations of the flat recipe

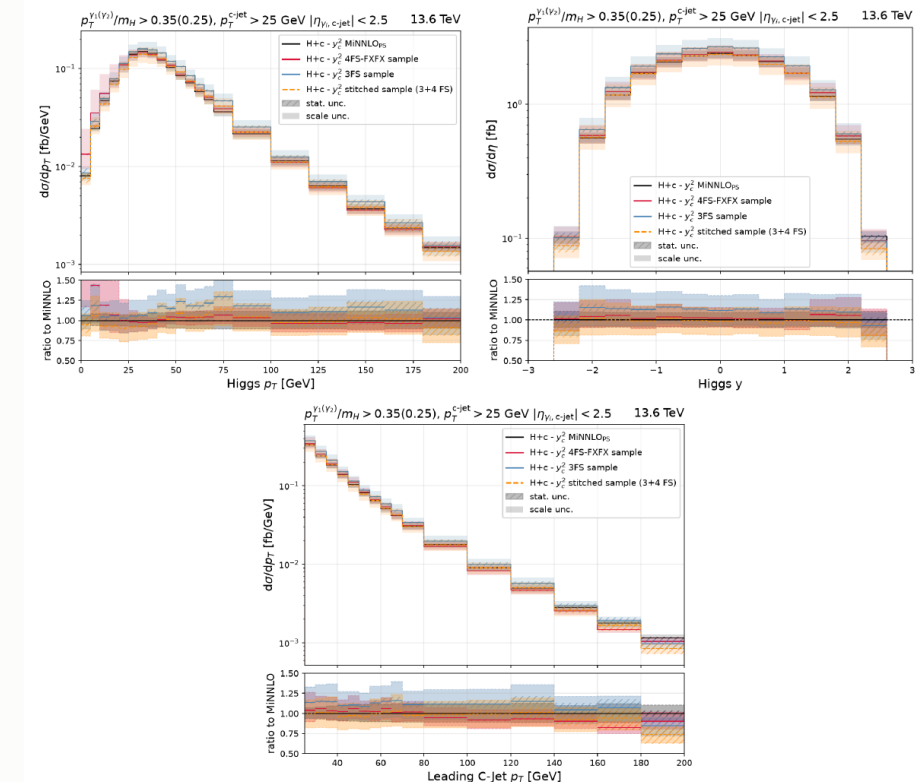
- compares 3FS with a 4FS sample (non-FxFx) that is **not** the nominal
- one number: no p_T dependence, no propagation through the selection
- **cannot be redone in Run 3**: there is no 3FS $H \rightarrow WW$ sample, only 3FS $H \rightarrow \gamma\gamma$

What LHCHWG-2026-007 tells us

Table 3 (13.6 TeV, $H \rightarrow \gamma\gamma$, $\gamma + c$ -jet fiducial, fb):

	3FS	4FS	3FS/4FS
NLO	0.0140	0.0226	0.62
NLO $\times$ own NNLO K	0.0182	0.0172	1.06

- the big plain-yield gap is mostly **normalisation** (perturbative order), not kinematics
- Fig. 11:** in a fiducial $\gamma\gamma + c$ -jet region, 4FS FxFx, 3FS, stitched and MiNNLO_{PS} agree in **shape** within ~ 10 – 15%
- the paper: the plain yield comparison "resulting in $O(30\%)$ discrepancy" is **not needed**



LHCHWG-2026-007, Fig. 11 (arXiv:2608.26863)

⇒ Evaluate the **scheme difference as a shape**, with both schemes at the same inclusive cross section.

Method

Event variable: leading GEN **c-hadron** p_T

- weakly-decaying open-charm hadron (D^0 , $D^\pm$, D_s , Λ_c , ...; charmonium excluded; $D^* \rightarrow D$ counted once)
- not from a b-hadron, $|\eta| < 2.5$, $p_T > T = \mathbf{5\ GeV}$ (10 GeV as check)
- stored in NanoAOD `GenPart` (Run 2 v9, Run 3 v13) → **same definition in derivation and in the analysis**
- **immune to the Higgs decay products** — GEN c-jets are not: photons / leptons are clustered into `GenJet s`

Weights — both schemes at the **same inclusive cross section** (pure fraction ratios, normalisation-free):

$$w(\text{bin}) = \frac{\sigma_{3FS}(\text{bin})/\sigma_{3FS}^{\text{incl}}}{\sigma_{4FS}(\text{bin})/\sigma_{4FS}^{\text{incl}}}, \quad w_0 \text{ for events with no c-hadron above } T$$

- **Up = w** (3FS-like), **Down = 2 - w** (mirror); nominal = 1
- inclusive cross section unchanged by construction; the variation **redistributes** events → changes the selection efficiency

Samples — all official, all files read

sample (H+c, MuRFScaleDynX0p50)	3FS files	3FS events	4FS FxFx files	4FS FxFx events
Run 3 2022preEE, $H \rightarrow \gamma\gamma$	54	1.06 M	61	2.20 M
Run 3 2022postEE, $H \rightarrow \gamma\gamma$	57	3.87 M	123	7.56 M
Run 3 2023, $H \rightarrow \gamma\gamma$	37	2.78 M	48	5.64 M
Run 3 2023BPix, $H \rightarrow \gamma\gamma$	28	1.37 M	27	2.68 M
Run 2 UL18, $H \rightarrow \gamma\gamma$	72	2.00 M	120	4.88 M
Run 2 UL18, $H \rightarrow WW \rightarrow 2\ell 2\nu$	4	5.00 M	68	19.8 M

- negative-weight fraction: **3FS $\approx 40\%$** , 4FS FxFx $\approx 21\%$
- Run 3 has **no 3FS $H \rightarrow WW$** $\rightarrow$ weights derived on $H \rightarrow \gamma\gamma$, transfer to $H \rightarrow WW$ validated in Run 2
- NanoAOD streamed over xrootd, 0 files skipped; standalone scripts, outside the analysis framework

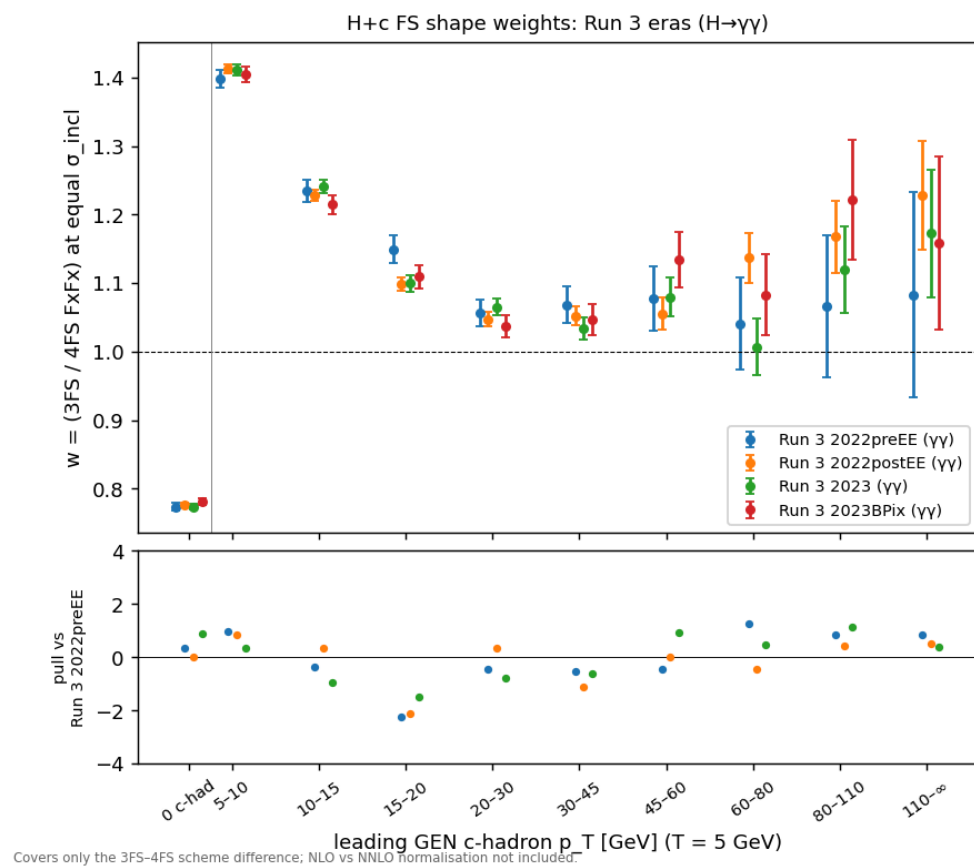
The weights — Run 3, four eras combined

leading c-hadron p_T [GeV]	none	5–10	10–15	15–20	20–30	30–45	45–60	60–80	80–110	>110
w_{up}	0.776	1.410	1.231	1.106	1.052	1.047	1.077	1.076	1.151	1.183
stat. unc.	0.002	0.004	0.005	0.007	0.007	0.009	0.016	0.023	0.035	0.051

- 3FS puts **more** of the cross section into events with a c-hadron above 5 GeV: $f(\geq 1c) = 0.595$ (3FS) vs 0.478 (4FS FxFx)
- largest effect at **low** c-hadron p_T (+41% at 5–10 GeV), 5–15% above 20 GeV
- delivered as correctionlib: `hplusc_fs_shape` (inputs `syst`, `gen_chad1_pt`; -1 if no c-hadron)

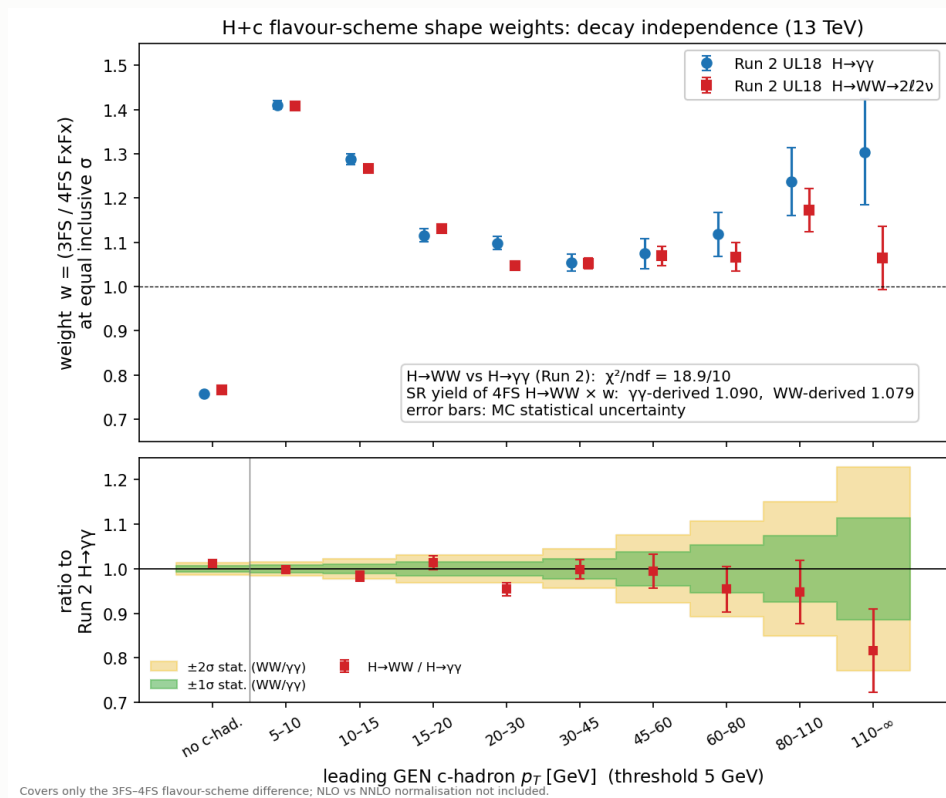
Consistent with the paper's Fig. 8: 3FS / 4FS FxFx leading D-hadron $p_T \approx 1.1$ – 1.25 at equal inclusive σ .

Validation 1 — Run 3 eras



All six era pairs compatible: $\chi^2/\text{ndf} = 6.2 - 10.0 / 10 \rightarrow$ one combined Run 3 table.

Validation 2 — $H \rightarrow \gamma\gamma$ vs $H \rightarrow WW$ (Run 2 UL18)

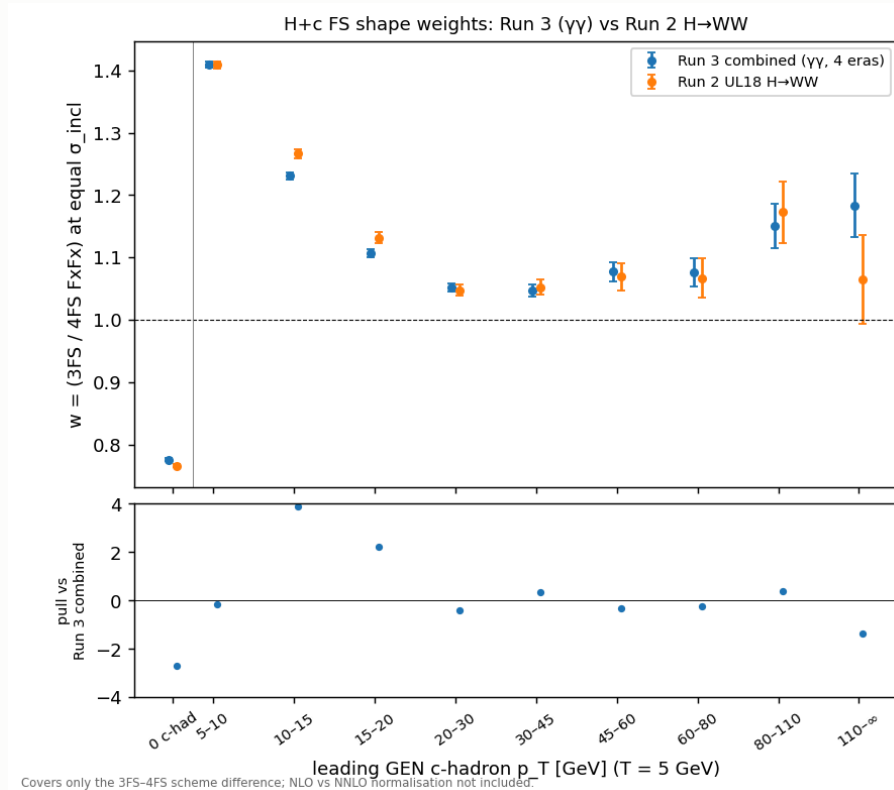


- ratio $H \rightarrow WW / H \rightarrow \gamma\gamma$ consistent with 1 within the **$\pm 1-2\sigma$ statistical bands** in 9 of 10 bins
- bins holding most of the cross section agree within **$\leq 1.6\%$**
- one bin at 2.9σ (20–30 GeV); $\chi^2/ndf = 18.9/10$
- weights applied to the 4FS $H \rightarrow WW$ sample, SR yield: $\gamma\gamma$ -derived **1.090** vs WW-derived **1.079** ($\sim 1\%$)

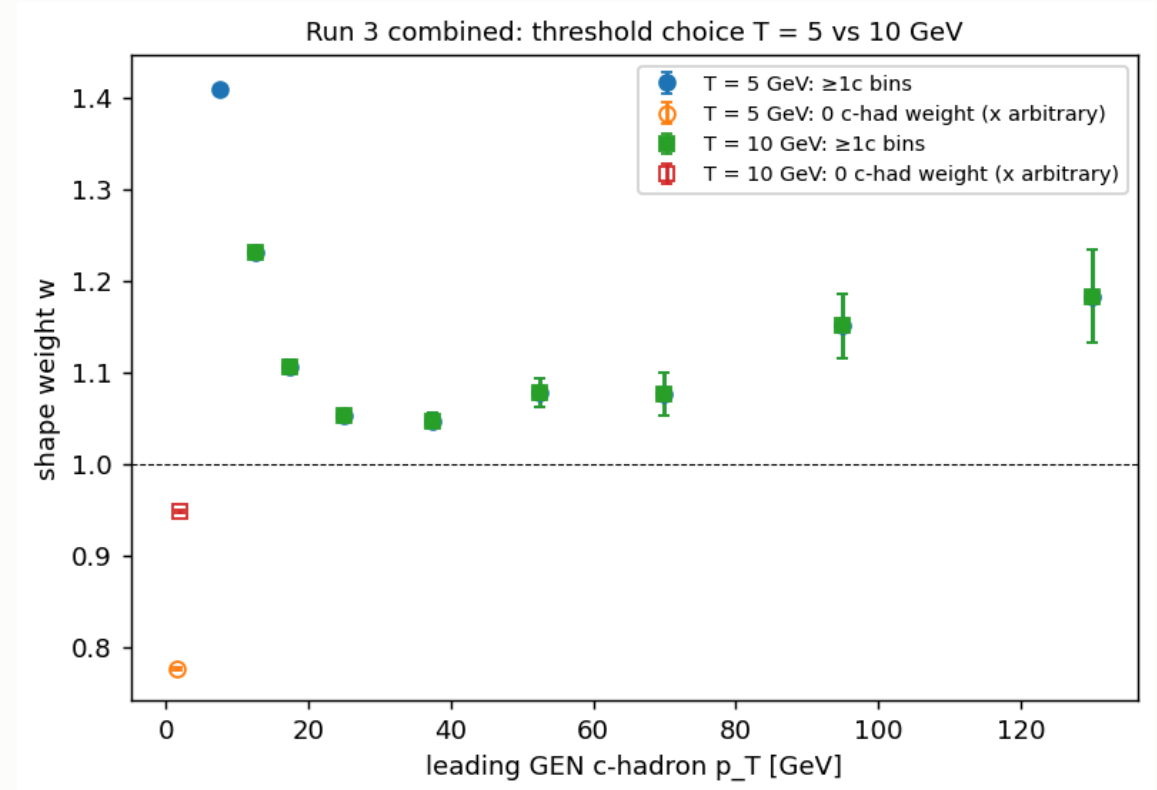
Verdict: weights derived on $H \rightarrow \gamma\gamma$ describe $H \rightarrow WW$ → the same procedure can be used in Run 3, where only 3FS $H \rightarrow \gamma\gamma$ exists.

The 20–30 GeV bin is a fluctuation of the Run 2 $H \rightarrow \gamma\gamma$ 3FS sample (backup).

Validation 3 — energy and threshold



Run 2 (13 TeV) vs Run 3 (13.6 TeV): differences **2–5%** (χ^2 30/10 for WW) — an energy effect; the Run 3 analysis uses the **Run 3** table.



$T = 5$ vs 10 GeV: **identical above 10 GeV** by construction; only w_0 changes ($0.78 \rightarrow 0.95$).

Reco-level validation on real $H \rightarrow WW$

Run 2 UL18 — the only era with a real 3FS $H \rightarrow WW$ sample

Standalone approximation of the selection: OS $e\mu$, tight μ / WP80 e , $p_T > 20/10$, $p_{T,\parallel} > 30$, $12 < m_{\parallel} \leq 72$, $m_{T,l2} > 30$, $m_{T,\parallel} > 60$; SR adds ≥ 1 jet $p_T > 20$ with DeepJet-medium c-tag (UL18 stand-in for PNet)

Closure on the SR yield

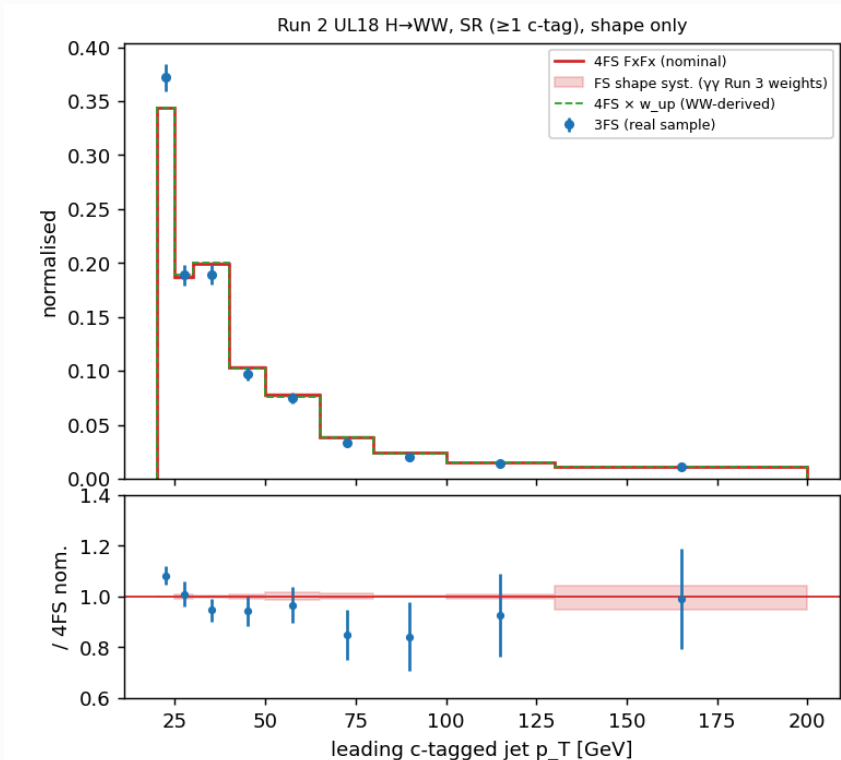
All samples at the **same inclusive cross section** (scheme comparison only):

region	real 3FS / 4FS	4FS $\times w_{\text{up}}$ ($\gamma\gamma$ Run 3)	($\gamma\gamma$ Run 2)	(WW-derived)	4FS $\times w_{\text{down}}$
preselection	1.041 ± 0.010	0.998	0.999	0.998	1.002
SR (≥ 1 c-tag)	1.063 ± 0.023	1.073	1.090	1.079	0.927

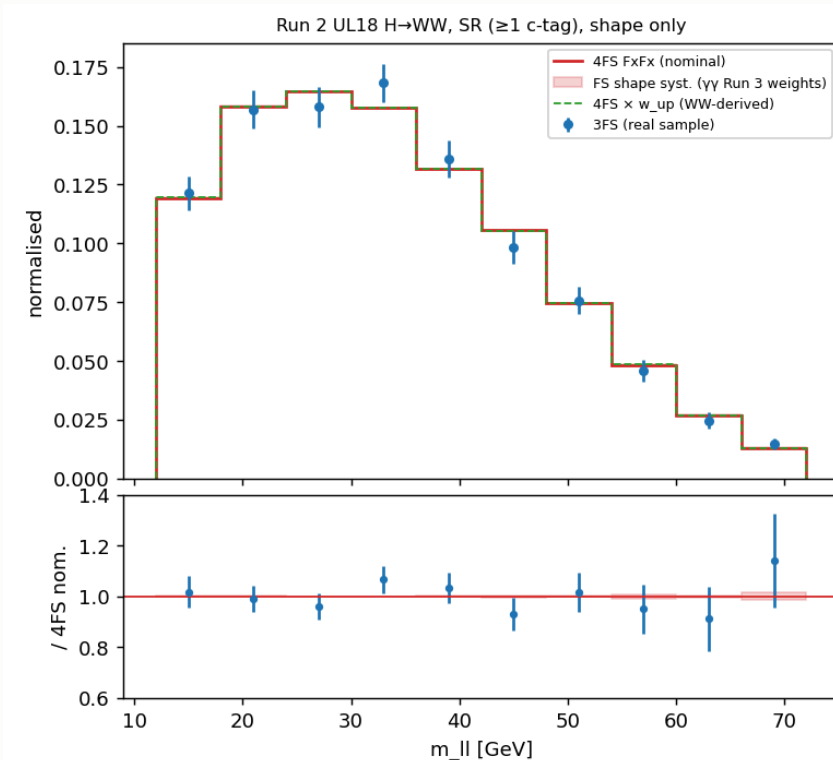
- the systematic moves the SR yield by **$\pm 7.3\%$** and **covers the real 3FS sample** ($+6.3 \pm 2.3\%$), closing within 0.5σ
- weights derived on $\gamma\gamma$ (Run 2 or Run 3) vs on WW itself: **$1.073 - 1.090$ vs 1.079** $\rightarrow$ decay transfer holds at reco level
- compare: Run 2 assigned a flat **30%**

Honest detail: part of the SR closure is a compensation — 3FS has +4% at preselection that the c-hadron weights do not model, while the weights give a larger c-tag gain (+7.5%) than the real 3FS (+2%).

SR shapes — 3FS vs 4FS FxFx with the systematic



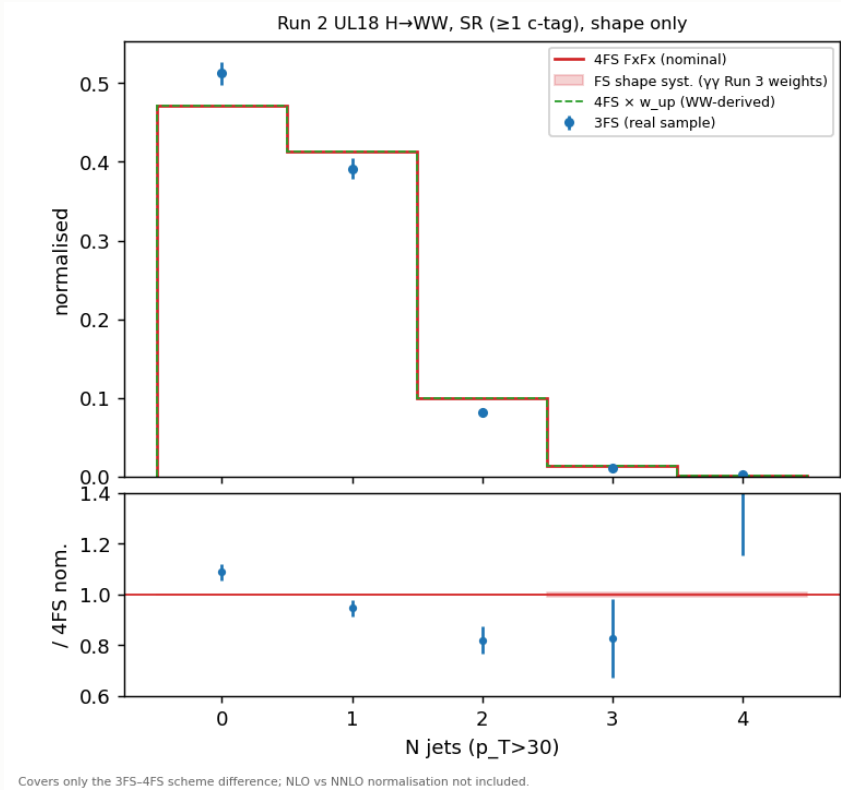
Covers only the 3FS–4FS scheme difference; NLO vs NNLO normalisation not included.



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3FS and 4FS FxFx agree within statistics in c-jet p_T/η , $m_{||}$, $p_{T,||}$, m_T , lepton p_T , MET, c-tag scores, N c-tags ($\chi^2 \approx \text{ndf}$). The systematic's shape effect in these observables is $\pm 1\text{--}4\%$.

Open item — jet multiplicity



N jets ($p_T > 30$)	4FS FxFx	3FS	3FS/4FS
0	0.471	0.512	1.09
1	0.413	0.391	0.95
2	0.100	0.082	0.82
3	0.014	0.011	0.8

- **$\chi^2 = 23/5$, not covered**: the c-hadron weights move N jets by only $\pm 1\%$
- partly a merging effect (FxFx has H+1j at ME level, 3FS is unmerged)

Options: (a) add an N-jet component derived from Run 2 H → WW; (b) check the N-jet weight in the fit first; (c) document as a limitation

Why not stitch — what it would cost

SR of the Run 2 $H \rightarrow WW$ study, same inclusive cross section:

	4FS FxFx (nominal)	stitched (3FS $\geq 1c$ + 4FS $0c$)	3FS
7-point scale envelope of the SR yield	+13.4 / -9.5%	+14.1 / -14.8%	+16.7 / -17.9%
negative-weight fraction	21%	mixed	40%
effective MC events in the SR (N_{eff})	55 971	dominated by 3FS part	2 265
3FS $H \rightarrow WW$ available in Run 3	—	no	no

- stitching needs 3FS $H \rightarrow WW$ samples that do not exist in Run 3, and brings a larger scale envelope and $\sim 25\times$ fewer effective events
- the shape systematic keeps the 4FS FxFx statistics and scale treatment, and reproduces the 3FS SR yield (slide 12)

Relation to LHCHWG-2026-007

paper	this approach
drop the plain massive-vs-massless yield uncertainty ($O(30\%)$)	✓ dropped
scheme differences are mostly normalisation; shapes agree within $\sim 10\text{--}15\%$ (Tab. 3, Fig. 11)	✓ evaluated as a shape, both schemes at equal inclusive σ
$\mu_R = \mu_F = H_T/4$, 7-point envelope	✓ samples use this setup; signal scale variations kept in the card
stitching when no NNLO+PS is available	— not done: no 3FS $H \rightarrow WW$ in Run 3; $\pm 15\%$ scale and $\sim 25\times$ MC-stat cost
split on c-jets with $p_T > 10$ GeV	— c-hadron $p_T > 5$ GeV (decay-immune, NanoAOD); identical above 10 GeV
"reweight the NLO+PS cross section with the most accurate inputs available"	— nominal kept at NLO — see the scope caveat

Summary

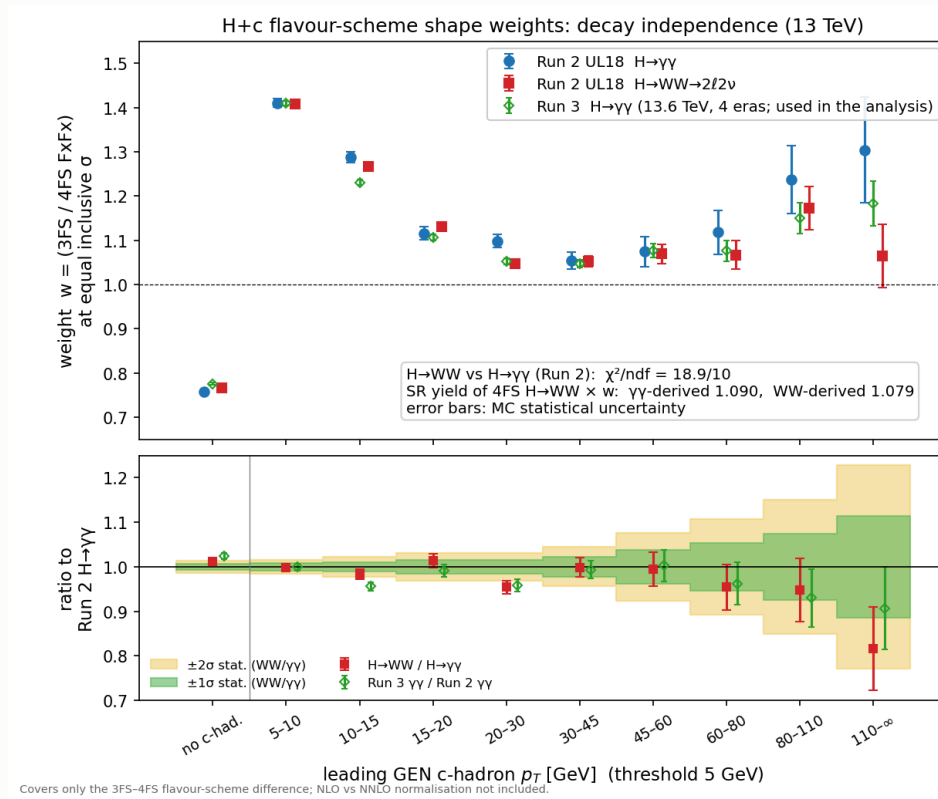
- The 30% `xsec_hplusc_4FS_5FS` lnN is replaced by a **3FS/4FS shape systematic** in leading GEN c-hadron p_T , with both schemes at the same inclusive cross section
- Derived on **official** Run 3 $H \rightarrow \gamma\gamma$ samples (four eras, consistent), delivered as `correctionlib`
- **Validated:** Run 3 eras ✓ · $H \rightarrow \gamma\gamma$ vs $H \rightarrow WW$ (1% on the SR yield) ✓ · real 3FS $H \rightarrow WW$ SR yield covered and closed within 0.5σ ✓
- Size on the SR yield: **$\pm 7.3\%$** vs the flat 30% before
- Open: **jet multiplicity** not covered (up to -18% at 2 jets) — decision needed

Scope: covers ONLY the 3FS–4FS flavour-scheme difference. It does not cover the NLO vs NNLO normalisation of the nominal sample ($\sim 25\%$ lower at NNLO, LHCHWG-2026-007 Table 3).

Next: GEN c-hadron column + weight in the framework → reprocess H+c signal → card: remove lnN, add `hplusc_fs_shape` → limit and impacts

Backup

Backup — the 20–30 GeV bin



- **Run 2 $H \rightarrow \gamma\gamma$ is the outlier, not $H \rightarrow WW$:** Run 3 $H \rightarrow \gamma\gamma$ (same decay) gives the same ratio to Run 2 $\gamma\gamma$ as $H \rightarrow WW$ does (0.955–0.96 at 20–30 GeV) → not a decay effect
- not a narrow spike: $\sim 5\%$ coherent excess of Run 2 $\gamma\gamma$ over 22–32 GeV (2 GeV bins, each $\leq 2.3\sigma$)
- no bad file: per-file χ^2 49/71 (3FS), 80/119 (4FS); jackknife-by-file error $\leq$ sum- w^2 error
- Run 2 $\gamma\gamma$ 3FS is the smallest sample: 2 M events, 40% negative weights
- $P(\geq 1 \text{ bin at } \geq 2.9\sigma \text{ among } 10) = 3.7\%$

→ **statistical fluctuation; the Run 3 weights used in the analysis agree with $H \rightarrow WW$ in this bin (1.052 vs 1.048).**

Backup — the $\gamma\gamma$ vs WW table ($T = 5$ GeV)

c-hadron p_T [GeV]	$H \rightarrow \gamma\gamma$	$H \rightarrow WW$	WW vs $\gamma\gamma$	pull
none	0.757 ± 0.005	0.766 ± 0.003	+1.2%	+1.5
5–10	1.410 ± 0.010	1.409 ± 0.006	−0.1%	−0.1
10–15	1.288 ± 0.012	1.266 ± 0.007	−1.7%	−1.6
15–20	1.116 ± 0.015	1.131 ± 0.009	+1.3%	+0.9
20–30	1.098 ± 0.015	1.048 ± 0.009	−4.6%	−2.9
30–45	1.054 ± 0.020	1.053 ± 0.013	−0.1%	0.0
45–60	1.074 ± 0.034	1.069 ± 0.022	−0.5%	−0.1
60–80	1.118 ± 0.050	1.067 ± 0.032	−4.6%	−0.9
80–110	1.237 ± 0.077	1.173 ± 0.049	−5.2%	−0.7
>110	1.304 ± 0.120	1.065 ± 0.072	−18%	−1.7

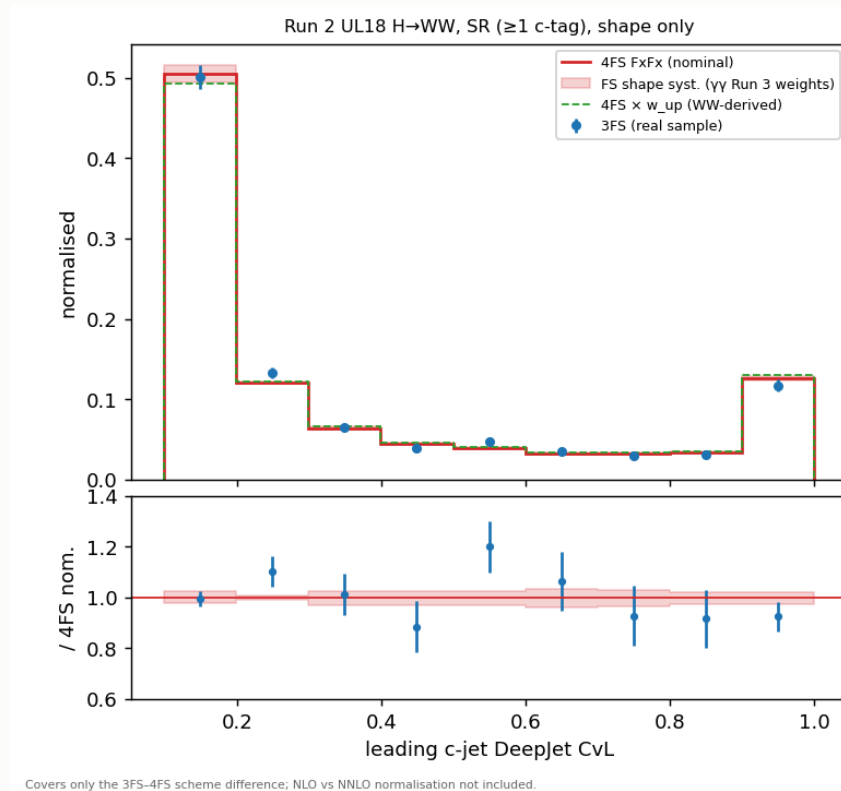
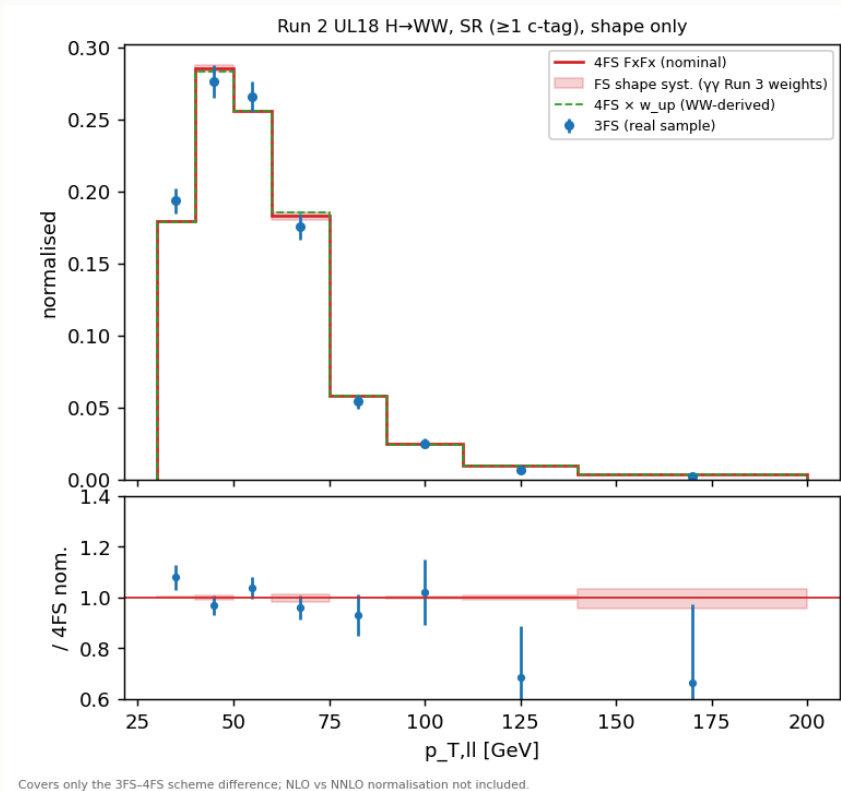
$T = 10$ GeV: $\chi^2 = 18.6/9$, same picture.

Backup — why a c-hadron and not a c-jet

definition	yy vs WW	usable in the framework
GEN c-jet (NanoAOD GenJet , hadronFlavour 4)	$\times$ $\chi^2 = 81/7$ — Higgs photons/leptons are clustered into the jets	yes
GEN c-jet, decay products removed by $\Delta R < 0.4$	$\sim$ shapes OK, 1.4% fraction offset (removes whole c-jets)	yes
GEN c-jet, Higgs constituents subtracted (MiniAOD)	$\checkmark$ R agrees to 0.4%	$\times$ needs MiniAOD
GEN c-hadron (NanoAOD GenPart)	$\checkmark$ SR yield agrees to $\sim 1\%$	$\checkmark$

- the official jet `hadronFlavour` is itself defined by ghost-associated c-hadrons $\rightarrow$ same charm content
- c-hadron $p_T \approx 60\%$ of the c-jet p_T : $T = 5 \text{ GeV} \approx$ the paper's 10 GeV c-jet split

Backup — reco SR: more observables



GEN leading c-hadron p_T in the SR: χ^2 82.9/10 (4FS nominal vs 3FS) → **7.0/10** after w_{up} — the weight does what it is built to do.

Backup — provenance

- scripts: `/eos/user/c/cgupta/HToWW/fs_unc/gen/` — `derive_chad.py`, `run_chad.sh`, `shape_weights.py`, `reco_dump.py`, `reco_ana.py`
- weights: `fs_unc/fs_shape/weights.{md,json}`, `correctionlib`
`fs_unc/fs_shape/hplusc_fs_shape_correctionlib.json`
- reco study tables: `fs_unc/fs_shape/reco_run2_WW.md`
- plots: `fs_unc/plots/fs_shape/` (GEN) and `fs_unc/plots/fs_shape/reco_run2_WW/` (reco)
- normalisation of the samples never enters: weights are ratios of fractions ($\sigma_{\text{bin}}/\sigma_{\text{incl}}$) of each scheme
- LHE scale weights: 9 per event in all samples, 7-point = indices 0,1,3,5,7,8
- Paper: LHCHWG-2026-007, arXiv:2608.26863 (§4.4, §5.3, Table 3, Figs. 8–11)